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Kathmandu, Nepal · Open to Remote Opportunities

Professional Summary

Backend Software Engineer with 3+ years of production experience building and owning distributed backend systems across government, education, and enterprise domains. Designed event-driven architectures handling concurrent workloads for platforms serving 1000+ users, and consistently delivered 20–30% performance gains through targeted query optimization and caching strategies. Known for translating ambiguous client requirements into well-scoped technical solutions, leading code reviews that raised team-wide quality standards, and mentoring junior engineers to independently own features. Seeking roles where backend architecture decisions have real product impact.

Technical Skills

Languages & Frameworks: Python, Django, Django REST Framework, FastAPI

Architecture & Concepts: RESTful API Design, Event-Driven Architecture, Distributed Systems, Async Processing, Concurrency, JWT Authentication, Role-Based Access Control

Databases & Caching: PostgreSQL (Query Optimization, Indexing, Schema Design), Redis (Caching, Pub/Sub)

Messaging & Task Processing: Celery, RabbitMQ, Redis Broker, Task Queues

DevOps & Tooling: Docker, CI/CD Pipelines, Git, GitHub, GitLab, Swagger/OpenAPI

Observability & Profiling: Django Silk, Django Debug Toolbar, Structured Logging

Deployment: Yeti Cloud, Himalayan Host (VPS), GCP, Linux-based server environments

Professional Experience

Backend Developer

July 2024 – Present

Calgen Nepal

Kathmandu, Nepal

- Owned the full backend lifecycle — requirements, architecture, implementation, deployment, and monitoring — across 6+ production systems spanning government portals, CRM platforms, and e-learning environments
- Reduced average PostgreSQL query execution time by **30%** through targeted indexing, query restructuring, and connection pool tuning on high-read CRM datasets
- Designed and implemented a multi-layer caching strategy using Redis, cutting repetitive DB hits and improving API response latency by an estimated **40%** on frequently accessed endpoints
- Architected distributed task pipelines with Celery + RabbitMQ/Redis supporting background job processing, real-time notifications, and scheduled reporting across platforms with concurrent users
- Served as the primary technical liaison with clients across multiple projects — gathering requirements, setting scope, communicating architectural trade-offs in non-technical terms, and ensuring delivery aligned with business goals
- Led code reviews across the engineering team, enforcing clean architecture, separation of concerns, and DRY principles — reducing review cycle time and improving merge quality
- Mentored 2 junior backend developers through structured pair programming and code review feedback, enabling them to independently own and ship features within 3 months
- Established structured logging, error handling frameworks, and monitoring standards adopted team-wide, improving incident detection and mean time to resolution

Django Developer (Internship)

October 2023 – January 2024

MindRisers Consortium

Kathmandu, Nepal

- Built and documented RESTful APIs using Django REST Framework under close mentorship, delivering features that passed production code review within the first month
- Designed normalized PostgreSQL schemas for relational data models and managed Docker-based dev environments for consistent team-wide builds
- Integrated third-party APIs and OAuth-based authentication systems, expanding platform security capabilities while coordinating requirements with external providers

Key Projects

RD Education Group — Education CRM & Student Portal | *Django DRF, PostgreSQL, Redis, RabbitMQ, Celery, Docker*
rdeducationgroup.com

- Architected a modular multi-tier CRM backend supporting agent dashboards, document workflows, and student lifecycle management — the most architecturally complex system in the portfolio
- Built a real-time cross-role chat and notification system using Redis Pub/Sub, eliminating the need for polling and reducing notification latency to near-instant delivery
- Designed JWT-secured API layer with granular multi-role access control, ensuring data isolation across student, agent, and admin tiers
- Automated deployments via CI/CD pipeline with Docker, achieving zero-downtime releases across multiple environments

Apex Education — Mock Test & School Management System | *Django DRF, PostgreSQL, Redis, Celery, Docker*

- Engineered a dynamic mock-test generation engine producing randomized, template-based question papers per student session — eliminating manual test preparation for staff
- Applied database normalization and strategic indexing to reduce query overhead on test-data reads under concurrent exam conditions
- Integrated Redis and RabbitMQ to decouple test-session processing from the request cycle, maintaining responsiveness under peak exam load

BodhiTreks.com — Travel E-Commerce Platform | *Django DRF, PostgreSQL, Redis, Celery, Docker* | bodhitreks.com

- Integrated SmartGateway third-party payment processing for secure online booking transactions, coordinating directly with the gateway provider on API contract and error handling
- Added SMS/OTP authentication, strengthening onboarding security without degrading user experience — validated with client team before release
- Optimized booking query performance on large CRM datasets; enhanced admin interface with django-jazzmin, improving internal operational efficiency

Laticrete CRM — B2B Product Survey System | *Django DRF, PostgreSQL, Redis, RabbitMQ, Celery, Docker*

- Built a regional B2B product survey platform for a multinational construction manufacturer (Africa, America, Nepal), implementing a complex hierarchical role-based API with region-scoped data access
- Translated layered organizational permission requirements from non-technical client stakeholders into a clean, maintainable RBAC implementation in Django
- Integrated real-time notifications and deadline reminders; tuned RabbitMQ message throughput to handle concurrent regional submissions without queue backlog

WWF Depository — Regional Data & Reporting Platform | *Django DRF, Django Templates, PostgreSQL, Redis, Celery* depository.wwfnepal.org.np

- Built a data entry and achievement reporting platform for **1000+ regional WWF employees**, leveraging Django Templates with HTML/CSS/JavaScript to deliver server-rendered dashboards, dynamic forms, and region-scoped access control
- Designed formatted CSV report generation for compliance and timely achievement reporting, working directly with WWF stakeholders to validate field selection and output format
- Optimized query performance on large employee datasets, ensuring report generation remained responsive at scale

RedPanda — Regional CRM & Reporting Platform | *Django DRF, PostgreSQL, Redis, Celery, Docker, HTML/CSS/JS*

- Delivered a data entry and reporting system for **1000+ staff** with region-assigned access, structured form workflows, and CSV report generation for operational tracking
- Managed end-to-end DevOps — server provisioning, deployment pipelines, and database administration — ensuring seamless platform availability and long-term maintainability

E.I.A / MOFE — Government Environmental Impact Portal | *Django DRF, PostgreSQL, Redis, Celery, Docker*

- Built a government portal for environmental impact assessment proposals, implementing proposal back-tracking across multiple ministries and a fine-grained role-permission system via Django Groups
- Developed an email and in-app notification system alerting ministry members and proponents on proposal status changes — requirements gathered and validated directly with government stakeholders
- Designed and optimized the database schema for multi-ministry proposal workflows; managed DevOps to ensure reliable availability for government users

ICAN VLE — Virtual Learning Environment | *Django DRF, PostgreSQL, Redis, Celery, Docker, HTML/CSS/JS*

- Built the study material upload pipeline for staff and contributed to in-browser PDF rendering with inline text commenting and highlighting — core to the student learning experience
- Integrated eSewa payment gateway for study material subscriptions, coordinating API contract and edge-case handling with the payment provider and client team
- Managed CI/CD and deployment pipelines maintaining **zero downtime** across releases, with proactive communication to the team during deployment windows

Impact & Leadership

Ownership: End-to-end delivery of 8 production systems across government, enterprise, and education sectors : from system design to post-release monitoring

Performance: Delivered 30% query time reduction and ~40% API latency improvement through systematic profiling and targeted optimization

Mentorship: Onboarded and mentored 2 junior engineers to feature-level independence within 3 months through structured code review and pair programming

Client Communication: Primary technical contact across multiple client-facing projects including translating business requirements into architecture and communicating trade-offs clearly to non-technical foreign as well as domestic clients

Team Standards: Introduced logging frameworks, review checklists, and CI/CD practices adopted across the engineering team

Education

Bachelor of Computer Applications (BCA)

Janamitri Multiple Campus

Kathmandu, Nepal

- Relevant Coursework: Data Structures & Algorithms, Database Management Systems, Software Engineering, Computer Networks, Object-Oriented Programming

Certifications

Python with Django – MindRisers Consortium

Django Developer Internship Certification – MindRisers Consortium

AI with Python – ProgrammingHub